not clear
would like to discuss it...

Transport = excitation transfer

# Optimal quantum transport on a ring via locally monitored chiral quantum walks


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1) FULL coherence might do worse

3) lift DARK states with CHIRALITY

OPTIMIZATION of det. probab. METHOD: 1. spectral P.-F. oper. 2. dark states

2) locally MONITORED continuous-time CHIRAL quantum walk

4) PVM monitoring without fine-tuning of the evolution time


In purely coherent transport on finite networks, destructive interference can significantly suppress transfer probabilities, which can only reach high values through careful fine-tuning of the evolution time or tailored initial-state preparations. We address this issue by investigating excitation transfer on a ring, modeling it as a locally monitored continuous-time chiral quantum walk. Chirality, introduced through time-reversal symmetry breaking, imparts a directional bias to the coherent dynamics and can lift dark states. Local monitoring, implemented via stroboscopic projective measurements at the target site, provides a practical detection protocol without requiring fine-tuning of the evolution time. By analyzing the interplay between chirality and measurement frequency, we identify optimal conditions for maximizing the asymptotic detection probability. The optimization of this transfer protocol relies on the spectral properties of the Perron-Frobenius operator, which capture the asymptotic non-unitary dynamics, and on the analysis of dark states. Our approach offers a general framework for enhancing quantum transport in monitored systems.


classical VS quantum

*Introduction.*—In a *classical random walk* the walker takes steps in random directions through a network, resulting in diffusive spreading. In a *quantum walk*, instead, the walker takes a quantum superposition of paths, leading to quantum interference effects [1, 2]. While constructive interference enables ballistic spreading (faster than the classical diffusive spreading), destructive interference can hinder transport or cause localization [3]. Continuous-time quantum walks are versatile for modeling coherent transport in discrete systems [4]. Equivalent to the tight-binding approximation in solid-state physics and in Hückel's molecular-orbital theory, their applicability is broad across systems [5–11]. Several studies on environment-assisted quantum transport have highlighted the beneficial interplay of coherent and incoherent dynamics [12–16], where the loss of phase coherence can enhance transport by suppressing destructive interference [17, 18]. Focusing on purely coherent transport, achieving unit transport efficiency in highly symmetric networks often requires non-trivial delocalized initial states due to symmetry-protected states [19, 20]. Breaking some symmetries, e.g., via network defects [21, 22], can improve transport.

1) constr. & destr. interf.

2) envir. assisted

3) SSB assisted

Chiral quantum walks leverage time-reversal symmetry breaking, typically through complex-valued edge weights [23, 24], to enable controlled directional transport [25–27] or its suppression [28], with promising applications in routing [29]. While chirality can mitigate destructive interference by affecting phase coherences, achieving nearly optimal transfer—even on a simple ring [26]—often requires fine control of the evolution time [30]. This represents an experimental challenge, as it requires a priori knowledge of the optimal time to halt the transfer protocol. Classically, this relates to first-passage-time problems [31–34], which address the statistics of the time a random walker takes to reach a target. However, in a quantum context the concept of first passage is not meaningful; instead, measurements must be explicitly incorporated [19], leading to the notion of first-*detected*-passage time [35–39]. The latter bridges unitary dynamics and measurement-induced collapse in monitored systems [40], offering insights into quantum-classical transition and measurement back-action. Although chirality has been shown to enhance transfer probabilities by lifting energy degeneracies that give rise to *dark states*—initial conditions which, suffering from destructive interference, prevent the detection of the desired state [38, 41, 42]—the optimal interplay between chirality and measurement frequency remains unexplored.

4) PTS breaking assisted

ROUTING

5) there is a BUT: evolution time

6) difficult meaning

...bridge

7) dark states (initial conditions)

8) CONTENT (interplay)

In this Letter, we address this gap by investigating a locally monitored chiral quantum walk on a ring (see Fig. 1), where the purely coherent evolution of the system is repeatedly interrupted by projective measurements at the target. First, we show the effective role of chirality and detection period in enhancing the detection probability at finite time. Then, we identify the optimal conditions for maximizing the (asymptotic) detection probability, deriving a general prescription that relies on the Perron-Frobenius spectrum of the non-unitary dynamics and on the analysis of dark states. Finally, we elaborate on the time scale for the asymptotic dynamics.

8.1

8.2

8.3

*Model and first-detected-passage-time problem.*—We investigate coherent excitation transfer on a cycle graph with $N$ sites (a ring) under local monitoring (see Fig. 1). The transfer is modeled as a continuous-time chiral quantum walk and the Hilbert space of the system is con-

1) CTCQW ring


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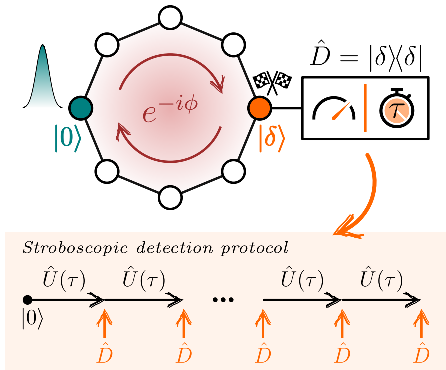


FIG. 1. Schematic illustration of the excitation transfer, modeled as a locally monitored chiral quantum walk under a stroboscopic detection protocol, investigated in this work.

span

veniently spanned by localized states at the sites of the graph, $\{|j\rangle\}_{j=0,\dots,N-1}$. In this basis, the Hamiltonian of the chiral quantum walk can be written as [25]

$$H = \sum_{j=0}^{N-1} e^{-i\phi}|(j+1)_N\rangle\langle j| + e^{i\phi}|(j-1)_N\rangle\langle j| \qquad (1)$$

modulo (PBC)

2) loop phase

3) circulant matrices

where $(j \pm 1)_N = (j \pm 1) \bmod N$ due to the periodic boundary conditions. We set all the diagonal elements to zero to solely focus on the effects of the phase $\phi \in \left[-\frac{\pi}{N}, \frac{\pi}{N}\right]$ responsible for the chirality. The upper bound follows from the fact that, in loops, the overall net phase is what matters, and so $-\pi \le N\phi \le \pi$ [26, 43]. Eigenvectors and eigenvalues of the Hamiltonian (1), a circulant matrix, respectively read

$$\boxed{|\lambda_j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i\frac{2\pi}{N}jk}\,|k\rangle\,, \quad \lambda_j = 2\cos\left(\phi - \frac{2\pi}{N}j\right), \quad (2)}$$

with $j = 0, \dots, N-1$. Under the time-independent Hamiltonian (1), the system initially prepared in a state $|\psi_0\rangle \equiv |\psi(t=0)\rangle$ evolves unitarily according to

$$|\psi(t)\rangle = \hat{U}(t)\,|\psi_0\rangle = e^{-iHt}\,|\psi_0\rangle\,, \qquad (3)$$

where we set $\hbar = 1$.

4) hitting time

stroboscopic measurement

In quantum systems the *first-detected-passage time*, or *hitting time*, is defined through repeated monitoring at the target site [19, 37], where the detector reveals the presence or absence of the walker. Here, we assume a stroboscopic detection protocol in which projective measurements are performed at times $\tau, 2\tau, 3\tau, \dots$, where the arbitrary finite detection period $\tau > 0$ is chosen by the experimentalist. The scheme is illustrated in the box in Fig. 1: starting from an initial state $|\psi_0\rangle = |0\rangle$ localized at a site (without loss of generality we assume the 0th), the monitored dynamics alternates unitary evolution, $\hat{U}(\tau)$ (3), and projective measurements at the target site, $\hat{D} = |\delta\rangle\langle\delta|$. In the following, we assume the target site to be the opposite to the initial one, i.e., $|\delta\rangle = |N/2\rangle$ for even $N$, and $|\delta\rangle = |(N \pm 1)/2\rangle$ for odd $N$. A typical run of the monitored dynamics produces a string of binary measurement outcomes: a sequence of *no* (the walker has not been detected) repeated until a *yes* (the walker has been detected) occurs at time $n\tau$, i.e., at the $n$th detection attempt. This time $n\tau$ is then defined as the first-detected-passage time for the run under investigation. The probability $F_n$ to first detect the walker at the $n$th attempt is $F_n = \langle\theta_n|\,\hat{D}\,|\theta_n\rangle$, where $|\theta_n\rangle = U(\tau)[(\mathbb{I} - \hat{D})U(\tau)]^{n-1}\,|\psi_0\rangle$ is the first-detection (unnormalized) vector [37]. The detection probability up to time $n\tau$ is

first det

probability unsuccessful = 1 - P —> compensates normalization

$$P_{\text{det}}(n) = \sum_{m=1}^{n} F_m. \qquad (4)$$

5) cumulative quantity

As a final remark, we point out that we investigate local monitoring for excitation transfer since purely-coherent transport performs poorly. Indeed, under unitary dynamics, the transfer probability $P_\delta(t) = |\langle\delta|\hat{U}(t)|\psi_0\rangle|^2$ typically exhibits low values with rare sharp peaks (see Sec. S1 of Supplemental Material (SM) [44]).

unitarily: rare sharp PEAKS

*Optimization problem.*—Our purpose is to leverage the key parameters of our model—the detection period $\tau$ between consecutive measurements, and the phase $\phi$ in the Hamiltonian (1)—to determine an optimal, robust transfer protocol: optimal, as it maximizes the detection probability (4) over $\phi$ and $\tau$, and robust, as it does not require fine-tuning of parameters. This optimization is subject to a constraint, as we assume the total time $T$ of the process to be a finite resource. Therefore, what we maximize is the detection probability $P_{\text{det}}(\phi, \tau)$ after $n$ detection attempts, where $n = \lfloor T/\tau \rfloor$, which is clearly upper bounded by its asymptotic value, $P_{\text{det}}(\phi, \tau) \le P_{\text{det}}(n \to \infty)$. In the following, we conveniently set $T = 200$, a value which allows us to achieve detection probabilities $> 90\%$ when the number of sites $N$ is relatively small (see the discussion on the asymptotic time scale at the end of the Letter), while keeping the number of measurements $n$ limited. Hereafter, we will focus first on the role of the phase $\phi$, and then on that of the detection period $\tau$.

1) what we can touch for OPTIMAL and ROBUST

2) total time is a FINITE resource

NO(?) Zeno

3) N FINITE as well (observe T and N)

*Optimal phase $\phi$.*—To gain initial insight into how the detection probability depends on the parameters of interest, Fig. 2 shows the density plots of $P_{\text{det}}(\phi, \tau)$ for a fixed number of sites, comparing even ($N = 20$) and odd ($N = 21$) cases. For even $N$ [Fig. 2(a)], we observe that for $\tau < \tau^*$ the detection probability is maximized at $\phi = 0$, decreases as $|\phi|$ increases, and vanishes at $\phi = \pm\pi/N$. Above this threshold $\tau^* \approx 1.55$ in the detection period, complex patterns emerge. Therefore, introducing a nonzero phase in the Hamiltonian (chirality) hinders excitation transfer between opposite sites of the symmetric cycle (even $N$), as it reduces the detection probability at the target. In contrast, chirality enhances this transfer in the asymmetric cycle (odd $N$) [Fig. 2(b)]. The directionality induced by the nonzero phases causes the detection probability to increase at either of the two target sites $\delta = (N \pm 1)/2$ opposite to

1) initial global view

2) even N: no phase works better

3) odd N: directionality fits well with imbalance

REPEAT

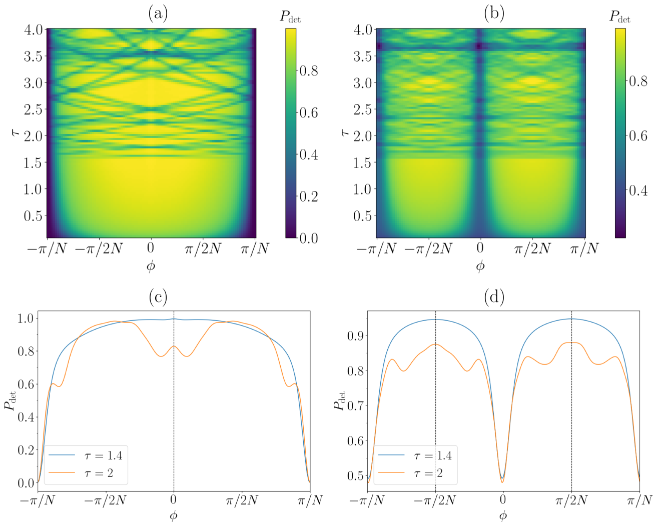


FIG. 2. Detection probability as a function of the phase $\phi$ and the detection period $\tau$ at fixed total observation time $T = 200$. Density plots $P_{\rm det}(\phi, \tau)$ (a) for $N = 20$ and $\delta = N/2$ and (b) for $N = 21$ and $\delta = (N-1)/2$. (c,d) Curves $P_{\rm det}(\phi)$ at given $\tau$ for system size $N$ and detection site $\delta$ as in (a,b), respectively.

REPEAT

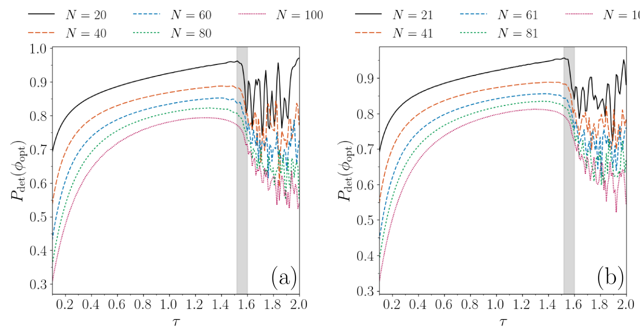


FIG. 3. Detection probability as a function of $\tau$ evaluated at fixed $\phi = \phi_{\rm opt}$ and $T = 200$ for various $N$. (a) Even $N$ with $\delta = N/2$ and $\phi_{\rm opt} = 0$. (b) Odd $N$ with $\delta = (N-1)/2$ and $\phi_{\rm opt} = \pi/2N$. For each $N$, a threshold $\tau^*$ can be identified within a relatively narrow region (grey-shaded).

the starting one. In particular, for $\tau < \tau^*$ the detection probability is always maximized at $\phi_{\rm opt} = \pm\pi/2N$, and minimized at $\phi = 0, \pm\pi/N$. Finally, we note that while $P_{\rm det}(\phi, \tau) = P_{\rm det}(-\phi, \tau)$ for even $N$, this symmetry is lifted for odd $N$ [Fig. 2(c,d)]. However, for odd $N$ the detection probability is symmetric upon changing both the target site *and* the sign of $\phi$.

*Optimal detection period $\tau$.*—The detection period $\tau$ plays a crucial role, as the detection probability for both even and odd $N$ shows two qualitatively distinct regimes separated by a sharp transition at a critical threshold $\tau^*$ [see Fig. 2(a,b)]. When $\tau < \tau^*$, $P_{\rm det}(\phi, \tau)$ is smooth, generally increasing with $\tau$ (recall we fixed $T = n\tau$), and remains high over extended intervals. Conversely, when $\tau > \tau^*$, a highly structured pattern with sharp oscillations emerges. As $\tau \to 0$, we observe a quantum Zeno effect [45, 46]. Here, frequent measurements performed on a state $|\delta\rangle$ (orthogonal to the initial one) effectively confine the walker's evolution in the subspace orthogonal to the detection subspace, preventing it from reaching $|\delta\rangle$. As a result, the detection probability approaches zero as $\tau \to 0$. A qualitatively analogous behavior is observed when varying the number of sites $N$, with the critical threshold $\tau^*$ exhibiting only minor variation. This is illustrated in Fig. 3, where $P_{\rm det}(\phi_{\rm opt}, \tau)$ is plotted for different $N$. As evident from Figs. 2-3, although maxima of the detection probability also occur in the region $\tau > \tau^*$, attaining them would require a fine tuning of $\tau$ that is difficult to achieve in practice. Since we are interested in a robust protocol, we maximize the detection probability in the relatively wide region $\tau < \tau^*$.

1) 2 regions

erratic oscill.

2) ring size N influence

threshold \tau seems N indep.

3) robust = NOT fine

Below, we present two complementary interpretations of the results above concerning the optimal parameters $\phi_{\rm opt}$ and $\tau_{\rm opt}$, and the threshold $\tau^*$: one based on the analysis of the Perron-Frobenius spectrum, and the other on the analysis of dark states.

4) interpretations

*Perron-Frobenius analysis.*—Local monitoring repeatedly interrupts the unitary evolution of the system by projective measurements, resulting in an overall non-unitary evolution. The elementary step—a free evolution (3) for time $\tau$ followed by the detection attempt $\hat{D} = |\delta\rangle\langle\delta|$—is governed by the (non-unitary) Perron–Frobenius operator [47]

1) overall non-unitary

$$\begin{aligned} \hat{O}(\phi, \tau) &= [\mathbb{I} - \hat{D}]\hat{U}(\phi, \tau) \\ &= \sum_j \mu_j(\phi, \tau)\, |\mu_j(\phi, \tau)\rangle\langle\mu_j(\phi, \tau)|\,. \end{aligned} \tag{5}$$

The last equality represents the eigendecomposition of the operator, which depends on both $\phi$ and $\tau$ via the evolution generated by the Hamiltonian (1). Hereafter, we will omit the explicit dependence of eigenvalues and eigenvectors on these parameters for shortness. The Perron-Frobenius analysis allows us to characterize the asymptotic dynamics of the system in the long-time limit [48–51]. Repeatedly applying such operator on an initial state $|\psi_0\rangle$ generates a non-unitary evolution which does not preserve the probability. The *survival probability*

2) eigen-decomp.

3) asympt.

$$S(n) = \|\hat{O}^n(\tau)|\psi_0\rangle\|^2 = \sum_{j=0}^{N-1} |\mu_j|^{2n} |\langle\mu_j|\psi_0\rangle|^2 \tag{6}$$

is the probability that the detector has not clicked after $n$ detection attempts—that is, the walker has not been detected yet [38]—and relates to the detection probability (4) via $S(n) = 1 - P_{\rm det}(n)$. The fact that $S(n) \leq 1$ means that $|\mu_j| \leq 1\ \forall j$. In particular, the contributions from eigenvectors with $|\mu_j| < 1$ vanish for $n \to \infty$, while those from eigenvectors with $|\mu_j| = 1$ survive. The asymptotic dynamics is thus determined by the largest-modulus eigenvalues and their corresponding eigenvectors. The analysis of the eigenproblem of $\hat{O}(\phi, \tau)$ will

4) contributions from ALL n powers... + prob. sum up to 1

5) tools for underst. (ODD N)

equip us with the necessary tools to understand both the presence of the threshold $\tau^*$ and the optimal values of the parameters. For clarity of discussion, we focus here on the case of odd $N$, deferring the case of even $N$ to Sec. S2 of SM [44].

6) threshold period emergence

The existence of the threshold $\tau^*$ is reflected in the Perron-Frobenius spectrum—evaluated at fixed $N$ and $\phi = \phi_{\rm opt}$—which differs between the regimes $\tau < \tau^*$ or $\tau > \tau^*$ [Fig. 4(a,b)]. For $0 < \tau < \tau^*$, all the eigenvalues fall within the unit circle, $|\mu_j| < 1\ \forall j$ [see also the largest-modulus eigenvalue in Fig. 4(d)], meaning that all the modes decay and so the excitation will be transferred to the target state with unit probability in the long-time limit, $P_{\rm det}(n \to \infty) = 1$. For $\tau > \tau^*$, instead, there may exist unit-modulus eigenvalues, indicating the emergence of modes which do not decay over time [52]. These results align with the numerical ones discussed above [Fig. 2(b,d)], including the estimate $\tau^* \approx 1.55$. The latter, in turn, represents the upper bound on the detection period, $\tau < \tau^*$, over which maximizing the detection probability to have a robust protocol.

modes that do not die out

7) RECIPE: minimize the largest modulus eigenvalue

The optimal parameters, $\tau_{\rm opt}$ and $\phi_{\rm opt}$, can be estimated by minimizing the Perron-Frobenius eigenvalue $\mu_{\rm PF}$ with the largest modulus, which governs the asymptotic dynamics in the long-time limit. The study of $|\mu_{\rm PF}|$ as a function of $\phi$ and $\tau$ in Fig. 4(c) reveals the following. At $\phi = \phi_{\rm opt} = \pm\pi/2N$ (odd $N$), $|\mu_{\rm PF}|$ decreases smoothly with increasing $\tau < \tau^*$, attains a minimum at $\tau_{\rm opt}$, and for $\tau > \tau^*$ its behavior becomes highly oscillatory, with $|\mu_{\rm PF}| = 1$ for specific values of $\tau$ [Fig. 4(d)]. At a given detection period $\tau < \tau^*$, $|\mu_{\rm PF}|$ is minimum at the optimal phase $\phi_{\rm opt}$ [Fig. 4(e)]. This behavior with respect to $\phi$ and $\tau$ is consistent with that of the detection probability in Fig. 2(b,d).

consistent

8) T is finite

A remark is in order. The Perron-Frobenius analysis characterizes the asymptotic behavior in the long-time limit, whereas previously we assumed a finite observation time $T$, which may not correspond to the long-time limit (we will discuss it at the end of the Letter). In this regard, letting $\tau_{\rm PF}$ be the value for which $|\mu_{\rm PF}|$ is minimum, we observe that $\tau_{\rm PF}$ is well approximated by $\tau_{\rm opt}$ for low $N$, and this agreement extends to higher $N$ upon increasing the total observation time $T$ (data not shown). In other words, the Perron-Frobenius analysis can be used to determine the optimal detection period, provided the total observation time is large enough. Below, we discuss the fact that $\tau_{\rm PF}$ approaches the value $\pi/2$ as $N$ increases [53], in terms of dark states.

consider scaling of N

9) \pi/2

1) splitting of the Hilbert space

*Dark state analysis.*—As thoroughly explained in [38], the presence of a detector in a finite-dimensional quantum system, such as ours, splits the Hilbert space into two parts: a *bright subspace*, whose states are guaranteed to be eventually detected, ($P_{\rm det}(\infty) = 1$), and a *dark subspace*, whose states are never detected, not even after an infinite number of detection attempts ($P_{\rm det}(\infty) = 0$). A state that lies in the dark subspace is orthogonal (and remains orthogonal under the action of the evolution operator) to the detection site $|\delta\rangle$. A state that satisfies these

2) orthogonal

DEPENDS ON THE MEASUREMENT PROCEDURE (SITE)

P.F. SPECTRUM
(eigenvalue relative to the success is 0)

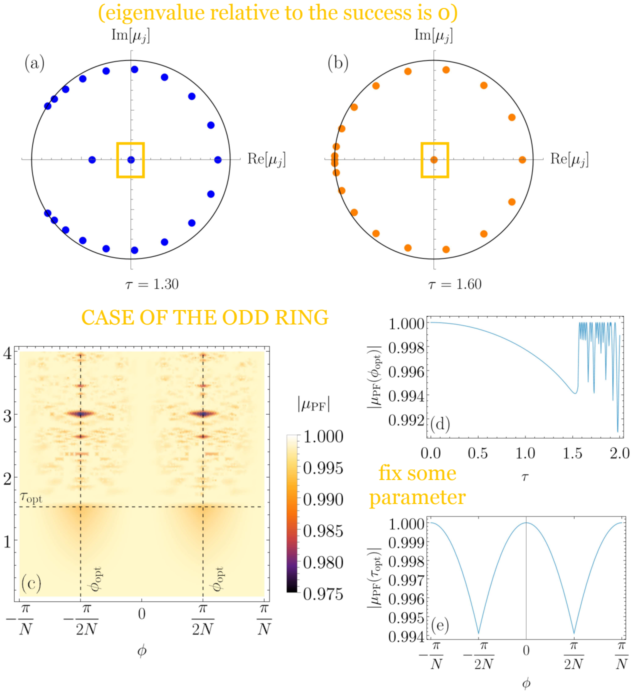


FIG. 4. Eigenvalues of the Perron-Frobenius operator (5) on the unit circle for two representative values of the detection period, (a) $\tau < \tau^*$ and (b) $\tau > \tau^*$ at $\phi = \pi/2N$, $\delta = 10$. (c) Density plot of the largest-modulus PF eigenvalue $|\mu_{\rm PF}|$ as a function of $\phi$ and $\tau$. (d) Curve $|\mu_{\rm PF}|$ as a function of $\tau$ at fixed $\phi = \pi/2N$. (e) Curve $|\mu_{\rm PF}|$ as a function of $\phi$ at fixed $\tau = 1.53$. Other parameters are: $N = 21$, $\delta = 10$.

3) definition of dark state: can also compute HOW DARK a state is

properties—i.e., it is stationary with respect to the evolution and detection attempts—is called a *dark state*. The total detection probability $P_{\rm det}(\infty)$ is the squared modulus of the projection of the initial state $|\psi_0\rangle$ on the *bright space* [38]. Accordingly, whenever $|\psi_0\rangle$ has a nonzero projection on the *dark space*, then $P_{\rm det}(\infty) < 1$, meaning that we are not guaranteed to detect the walker even with infinite measurements. Understanding when dark states arise and characterizing them is therefore important to optimize the detection probability.

4) build dark states from 2 energy eigenstates

Considering the time evolution in Eq. (3), a dark state can be built from the energy eigenstates (2),

$$|\gamma_{n,m}\rangle = \sqrt{\frac{N}{2}}\Big(\langle\delta|\lambda_n\rangle|\lambda_m\rangle - \langle\delta|\lambda_m\rangle|\lambda_n\rangle\Big), \qquad (7)$$

with $m \neq n$. Letting it evolve with $\hat{U}(\tau)$, we get the state

$$\hat{U}(\tau)|\gamma_{n,m}\rangle = \sqrt{\frac{N}{2}}\Big(e^{-i\lambda_m\tau}\langle\delta|\lambda_n\rangle|\lambda_m\rangle - e^{-i\lambda_n\tau}\langle\delta|\lambda_m\rangle|\lambda_n\rangle\Big), \qquad (8)$$

trick lies in the antisymmetric shape

which is orthogonal to the detection site $|\delta\rangle$ if

$$\lambda_m\tau \equiv \lambda_n\tau \pmod{2\pi}, \qquad (9)$$

in which case the system never visits $|\delta\rangle$, resulting in $P_{\rm det}(\infty) = 0$. If no pairs of eigenvalues $\lambda_m, \lambda_n$ satisfy

5) condition to be fully bright

NOTE: use orthog. + phase that factors out

the condition (9) and $\langle\lambda_j|\delta\rangle \neq 0\ \forall j$, then the Hilbert space is *fully bright*, and there are no dark states [38]. Note that when Eq. (9) holds true, we have that $[\mathbb{I} - \hat{D}]\hat{U}(\tau)|\gamma_{n,m}\rangle = e^{-i\lambda_m\tau}|\gamma_{n,m}\rangle$, i.e., these dark states are eigenstates of the Perron-Frobenius operator $\hat{O}$ and are the only ones with unit-modulus eigenvalue [54]. In our model the condition $\langle\lambda_j|\delta\rangle \neq 0\ \forall j$ is satisfied by any localized state $|\delta\rangle$ [see Eq. (2)], as our detection states, while the condition (9) can be satisfied in two ways:

6) localized states have components along every energy eigenst.

7) 2 ways condition (9) can be satisfied

(i) When the Hamiltonian's spectrum is degenerate, then Eq. (9) is satisfied $\forall\tau$ for each degenerate level. Degeneracies in the spectrum (2) arise only for $\phi = 0$ and $\phi = \pm\pi/N$, and the properties of the resulting dark states depend on the parity of $N$.

symmetric band!

For even $N$, when $\phi = 0$ there exist $\frac{N}{2} - 1$ two-fold degenerate levels. In this case, all the stationary dark states constructed from the pairs of eigenstates in each degenerate level are orthogonal to the initial state. Therefore, they neither contribute to the dynamics of the system nor affect the detection probability. When $\phi = \pm\pi/N$ there exist $N/2$ two-fold degenerate levels. It can be proved (see Sec. S2 of SM [44]) that the initial state lies in the dark subspace spanned by the stationary dark states constructed from these degenerate levels. As a consequence, in this case the walker is never detected in $|\delta\rangle$, yielding $P_{\rm det}(\pm\pi/N) = 0$ in Fig. 2(a).

7.1) degeneracy

For odd $N$, calculations analogous to those in Sec. S2 of SM [44] reveal that for $\phi = 0, \pm\pi/N$ the total overlap of the initial state with the dark space is 1/2, thus explaining the minima of the detection probability for these values of $\phi$ in Fig. 2(b). For such values of $\phi$, there exist $(N-1)/2$ two-fold degenerate levels, each contributing a dark state that, together, form the basis spanning the dark space.

7.2) combinations

(ii) Using any nondegenerate couple of eigenvalues $\lambda_m \neq \lambda_n$, there may exist proper combinations of $\tau$ and $\phi$ that lead to $\lambda_m\tau = \lambda_n\tau + 2k\pi$ for some $k \in \mathbb{Z}$. Plugging the eigenvalues (2) in the latter expression, i.e., in Eq. (9), we derive the condition relating $\tau$ and $\phi$ under which dark states appear

condition

$$\boxed{\tau = \frac{k\pi}{2\sin\left(\frac{\pi(m-n)}{N}\right)\sin\left(\phi - \frac{\pi(m+n)}{N}\right)}}, \qquad (10)$$

which establishes the lower bound $\tau \geq \pi/2$ for the presence of dark states arising from nondegenerate levels.

8) large N limit

maximize sines

Such a lower bound is exactly attained for even $N$ when $\phi = 0$. In all other cases, the $\tau$ at which the first dark state appears approaches $\pi/2$ as $N$ grows. This is because the discrete arguments of the sine functions in Eq. (10) become denser in the interval $[0, 2\pi)$ for increasing $N$. For large $N$, it is therefore possible to find pairs $(m, n)$ such that the product of the two sine functions, regardless of $\phi$, is arbitrarily close to 1, implying the condition $\tau \gtrsim \pi/2$. In other words, for any $\phi$ the first dark state associated with nondegenerate levels will arise at $\tau \to \pi/2$ as $N \to \infty$, appearing as a dark-state line at $\tau \to \pi/2$ in the parameter space $(\phi, \tau)$. In general, for finite $N$, solutions to Eq. (10) constitute dark-state curves in the $(\phi, \tau)$-space, which correspond to low-detection-probability curves. Figures 2(a,b) represent a clear example: The detection probability in the parameter space $(\phi, \tau)$ is low along the curves associated with the presence of dark states, including a nearly straight line at $\tau \gtrsim \pi/2$. In the region $\tau \geq \pi/2$, the dark-state curves become denser as $N$ increases, because the number of dark states from nondegenerate levels is of order $O(N^2)$, and also as $\tau$ increases (see Sec. S3 of SM [44]).

9) dark state lines

To summarize, in the region $\tau < \pi/2$, for even $N$ dark states at $\phi = 0$ do not affect the evolution of $|\psi_0\rangle$ yielding $P_{\rm det}(\infty) = 1$, while for $\phi = \pm\pi/N$ the initial state is a dark state, yielding $P_{\rm det}(\infty) = 0$; for odd $N$, the initial state has a finite overlap with dark states at $\phi = 0, \pm\pi/N$, yielding $P_{\rm det}(\infty) = 1/2$. For $\phi \neq 0, \pm\pi/N$, the Hilbert space is fully bright in this region, so $P_{\rm det}(\infty) = 1$ for both even and odd $N$. On the other hand, in the region $\tau \geq \pi/2$, the number of dark states increases as $N$ and $\tau$ increase, yielding low-detection-probability curves in the $(\phi, \tau)$-space.

*Time scale of the asymptotic dynamics.*—As pointed out above, the Perron-Frobenius analysis provides useful insights into the asymptotic dynamics of the system. Related to our assumption of a finite observation time, this naturally leads to the question: What is the time scale over which such asymptotic behavior emerges? Such a time scale is governed by the spectral properties of the Perron-Frobenius operator (5), in particular by the spectral gap $\Delta \equiv 1 - |\mu_{\rm PF}|$, where we recall $\mu_{\rm PF}$ is the largest-modulus eigenvalue, according to which the asymptotic dynamics time scale can be estimated as $t_{\rm as} \sim \Delta^{-1}$. Because the Perron-Frobenius operator depends on the detection period $\tau$, the asymptotic time scale $t_{\rm as}$ varies with $\tau$ as well, as shown in Fig. 5(a). In the limit of small $\tau$, $t_{\rm as}$ diverges due to the quantum Zeno effect, which prevents the walker from reaching—and thus being detected at—the measurement site. As $\tau$ increases, $t_{\rm as}$ monotonically decreases, reaching its minimum at $\tau_{\rm opt}$ (minimum of $|\mu_{\rm PF}|$), after which it quickly increases. Upon crossing the threshold $\tau^*$, the system transitions to a regime characterized by a high sensitivity to small variations in the model parameters $\tau$ and $\phi$ [Fig. 2], a sensitivity also reflected in the values of $t_{\rm as}$ [Fig. 5(a)]. This behavior is consistent upon varying the number of sites $N$, with larger $N$ leading to higher values of $t_{\rm as}$. The fact that the $\tau_{\rm opt}$ minimizing $|\mu_{\rm PF}|$ approaches $\pi/2$ as $N$ increases is reflected in the corresponding behavior of the minimum of $t_{\rm as}$ in Fig. 5(a).

1) finite T

2) spectral properties: GAP

decay time like energy^-1

3) timescale depends on \tau

4) scaling with N

The asymptotic time at which the detection probability saturates increases as $N$ increases. However, in practical cases, the experimentalist may be satisfied with the detection probability exceeding a certain threshold. This motivates the study in Fig. 5(b), where $P_{\rm det}$ is shown as a function of $N$ and the total observation time $T$ at $\tau_{\rm opt}$ and $\phi_{\rm opt}$ derived from the Perron-Frobenius analysis. First, we observe that the larger $N$, the larger $T$ at which $P_{\rm det}$ saturates (i.e., $T \sim t_{\rm as}$), and the larger the

5) reach threshold

intuitively go well together...

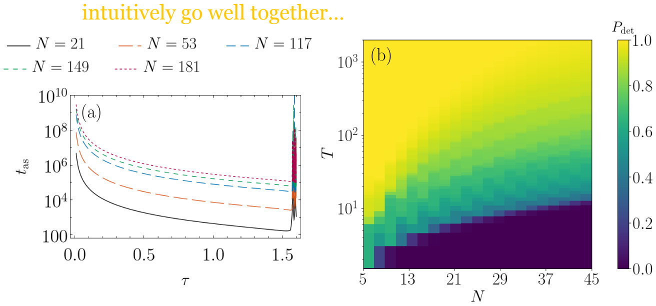


FIG. 5. (a) Asymptotic time $t_{as}$ as a function of the detection period $\tau$ for various odd $N$. (b) Density plot of $P_{det}$ as a function of odd $N$ and total observation time $T$, evaluated at the $\tau_{opt}$ corresponding to each $N$ and at $\delta = (N-1)/2$. In both panels the phase is fixed at $\phi_{opt} = \pi/2N$.

minimum $T$ at which $P_{det}$ is non-vanishing. Second, we note that the observation time $T$ required to reach, e.g., $P_{det} \approx 80\%$ is almost an order of magnitude lower than that required to saturate. Third, we note that the observation time $T = 200$ considered in the present work is sufficiently long to observe the asymptotic dynamics and the saturation of $P_{det}$ for $N < 21$, with $T \sim t_{as}$ at $N = 21$. For larger $N$, the system takes longer to enter the asymptotic regime. Provided that the observation time satisfies $T \gtrsim t_{as}$, then $P_{det}$ can saturate and the optimal detection period $\tau_{opt}$ is identified by the minimum of $|\mu_{PF}|$. Instead, when $T < t_{as}$, the optimal detection period $\tau_{opt}$ differs from that determined from the Perron-Frobenius analysis (minimum of $|\mu_{PF}|$), say $\tau_{opt}^{(PF)}$, and it decreases with $N$ at a rate that depends on $T$: the longer $T$, the slower the decrease, with $\tau_{opt}$ approaching $\tau_{opt}^{(PF)}$ as $T \to \infty$ (data not shown).

*Conclusions.*—In this Letter, we have investigated how chirality and local monitoring jointly enhance excitation transfer, modeled as a continuous-time quantum walk on a ring. We show that optimizing both the chiral phase $\phi$ and the detection period $\tau$ overcomes limitations of purely unitary dynamics, such as destructive interference and dark states. The resulting transfer protocol is both

phase and detection period make it better

optimal and robust

optimal, yielding significantly higher detection probabilities, and robust, requiring no fine-tuning of parameters or tailored initial-state preparations. Our approach combines two key insights: (i) The identification of dynamically relevant dark states (see Sec. S2 of SM [44]), and (ii) the spectral analysis of the non-unitary Perron-Frobenius operator to determine optimal parameters. This provides a general framework for improving transport in monitored quantum systems.

2 key steps:

Beyond advancing the understanding of first-detected-passage problems in closed loops, our results offer practical guidelines that can be readily extended beyond quantum transport to the design of efficient quantum protocols in domains where destructive interference and dark states usually limit performance, such as search algorithms [55, 56], state transfer [57], quantum routers [29], and potentially in quantum-walk-based computing [58].

fight against destr. interf. and dark states

Finally, hitting times play a central role in the performance of quantum protocols, as they directly affect how rapidly a target state can be reached or detected. The recent implementation of local monitoring via mid-circuit measurements—either strong [41, 59] or weak [60]—demonstrates the growing experimental feasibility of accessing quantum hitting-time statistics on current quantum hardware. In parallel, restart strategies have emerged as a new, powerful control mechanism to further speed up hitting times in monitored quantum systems [42, 61–63]. Together, these advances underscore the relevance of our approach for designing efficient and experimentally viable quantum protocols.

hitting times

?????

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Hitting time DEFINIT.

Localization

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[53] This trend is not captured by Fig. 3, where the maximum of the detection probability shifts instead toward lower $\tau$, due the finite observation time $T$.
[54] The time-evolution operator $\hat{U}(\tau)$ in Eq. (3) is unitary, thus preserves the norm and has unit-modulus eigenvalues. Accordingly, for the Perron-Frobenius operator (5) to have an eigenvalue of modulus 1 with corresponding eigenstate $|\mu\rangle$, it must be that $\hat{D}\hat{U}(\tau)|\mu\rangle = 0$; otherwise, the non-unitary operator $(\mathbb{I} - \hat{D})$ would reduce the norm of $\hat{U}(\tau)|\mu\rangle$, preventing a unit-modulus eigenvalue of $\hat{O}$. In other words, $|\mu\rangle$ is eigenstate of $\hat{O}$ with unit-modulus eigenvalue only if $\hat{O}|\mu\rangle = \hat{U}(\tau)|\mu\rangle$, meaning that $|\mu\rangle$ must be an eigenstate of $\hat{U}(\tau)$ orthogonal to $|\delta\rangle$.
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# Supplemental Material for
# "Optimal quantum transport on a ring via locally monitored chiral quantum walks"


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(Dated: July 18, 2025)



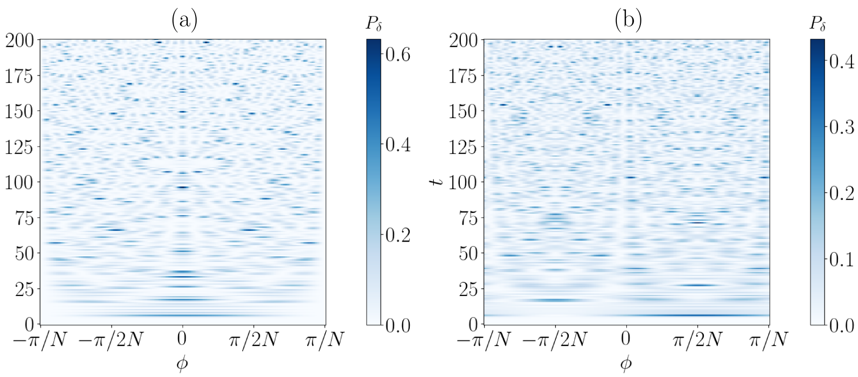
This Supplemental Material provides additional information on the results discussed in the main text. Here, section, equation, and figure numbers are prefixed with "S"; numbers without this prefix refer to those in the main text.


## S1. PURELY-COHERENT TRANSPORT UNDER UNITARY EVOLUTION

narrow peaks

As anticipated in the main text, the purely-coherent transport under unitary dynamics is not efficient. The instantaneous probability $P_\delta(\phi,t) = |\langle\delta|\hat{U}(\phi,t)|\psi_0\rangle|^2$ typically remains low throughout the evolution, punctuated by narrow, sharp peaks. This is evident from the comparison between Fig. 2 and Fig. S1: Fig. S1 shows $P_\delta$ as a function of the phase $\phi$ and time $t$, revealing that its values are consistently and significantly lower over the entire parameter range considered than those of the detection probability $P_{\rm det}$ in Fig. 2.

even




odd

toward localized site

SSB

FIG. S1. Instantaneous transfer probability $P_\delta$ from site 0 to $\delta$ as a function of the phase $\phi$ and time $t$ (a) for $N = 20$ and $\delta = N/2$ and (b) for $N = 21$ and $\delta = (N-1)/2$.

## S2. PERRON-FROBENIUS ANALYSIS FOR EVEN $N$

odd

here even

The Perron-Frobenius analysis in the main text focused on the case of odd $N$. Here we discuss the case of even $N$, which needs to be considered more carefully. The main differences between even and odd $N$ root back to the degeneracies of the energy spectrum in Eq. (2). For odd $N$, all the energy levels but one—the highest ($j = 0$)—have a two-fold degeneracy when $\phi = 0$. These degeneracies are lifted when $\phi \neq 0$. For even $N$, instead, all the energy levels but two—the lowest ($j = N/2$) and the highest ($j = 0$)—have a two-fold degeneracy when $\phi = 0$. Nonzero phases lift these degeneracies except when $\phi = \pm\pi/N$, in which case all the energy levels, including the lowest and the highest, have a two-fold degeneracy. Following [1], we can construct a dark state from each degenerate level $\lambda_j$ as follows

two-fold degeneracy can be seen on the unit circle EASILY

$$\boxed{|\gamma_j\rangle = \sqrt{\frac{N}{2}}\Big(\langle\delta|\lambda_{j,2}\rangle|\lambda_{j,1}\rangle - \langle\delta|\lambda_{j,1}\rangle|\lambda_{j,2}\rangle\Big).} \tag{S1}$$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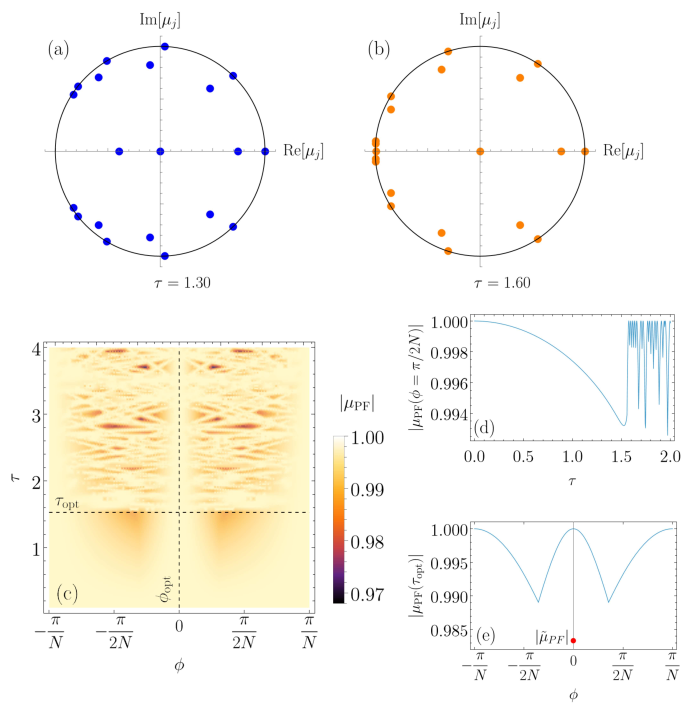


what is tau star
subleading eigenvalue

FIG. S2. Eigenvalues of the Perron-Frobenius operator (5) on the unit circle for two representative values of the detection period, (a) $\tau < \tau^{*}$ and (b) $\tau > \tau^{*}$ at $\phi = 0$. (c) Density plot of the largest-modulus PF eigenvalue $|\mu_{\rm PF}|$ as a function of $\phi$ and $\tau$. (d) Curve $|\mu_{\rm PF}|$ as a function of $\tau$ at fixed $\phi = \pi/2N$. (e) Curve $|\mu_{\rm PF}|$ as a function of $\phi$ at fixed $\tau = 1.53$. The red circle denotes the subleading PF eigenvalue $|\tilde{\mu}_{\rm PF}|$, which is the relevant one for the asymptotic dynamics at $\phi = 0$ (the initial state has zero overlap with dark states). Other parameters are: $N = 20$, $\delta = 10$.

DARK STATES

For $\phi = 0$, the dark states read [1]

$$|\gamma_j\rangle = \sqrt{\frac{2}{N}} \sum_{k=0}^{N-1} \sin\left(\frac{2\pi jk}{N}\right)|k\rangle \tag{S2}$$

NO DARK COMPONENT

irrelevant

with $j = 1, \ldots, N/2 - 1$, and have no overlap with the initial state, $\langle 0|\gamma_j\rangle = 0\, \forall j$. As a result, these states do not contribute to the walker's dynamics and can be considered irrelevant in the evolution. These dark states are eigenstates of the Perron-Frobenius operator with unit-modulus eigenvalue [see Fig. S2(a,b)]. Consequently, the asymptotic dynamics relevant to excitation transfer will be governed by the subleading eigenvalue, defined as $\tilde{\mu}_{\rm PF} \equiv \max_{|\mu_k|<1}(\mu_k)$, i.e., the largest eigenvalue in modulus strictly less than one.

For $\phi = -\pi/N$, instead, the dark states read

$$|\gamma_j\rangle = \frac{1}{\sqrt{2N}} \sum_{k=0}^{N-1} \left(e^{i\frac{2\pi jk}{N}} + e^{-i\frac{2\pi(j+1)k}{N}}\right)|k\rangle \tag{S3}$$

MAXIMAL DARKNESS

relevant: ALWAYS overlap

with $j = 0, \ldots, N/2 - 1$, and always overlap with the initial state, $\langle 0|\gamma_j\rangle = \sqrt{2/N}\, \forall j$. The initial state turns out to be completely dark for $\phi = -\pi/N$, as it is the equal superposition of all the $N/2$ dark states forming a basis for the dark subspace, $|0\rangle = \sqrt{2/N}\sum_{j=0}^{\frac{N}{2}-1}|\gamma_j\rangle$. The case $\phi = \pi/N$ can be treated analogously. Therefore, $P_{\rm det}(n \to \infty) = 1$ if $\phi = 0$, and $P_{\rm det}(n \to \infty) = 0$ if $\phi = \pm\pi/N$ [2].

DARK subspace

OPTIMAL tau

The study of the largest-modulus Perron-Frobenius eigenvalue $\mu_{\rm PF}$ is shown in Fig. S2(c). Similarly to the case of odd $N$, $|\mu_{\rm PF}|$ smoothly attains a local minimum (which we identify as $\tau_{\rm opt}$) in the interval $\tau < \tau^{\star}$, and then rapidly oscillates [Fig. S2(d)]. As for the dependence on the phase $\phi$, $\mu_{\rm PF}$ shows two minima at $\phi \neq 0$, in contrast with the optimal phase $\phi_{\rm opt} = 0$ we expect for even $N$ from the numerical optimization reported in the main text. On the other hand, as discussed above, dark states built using degenerate energy levels are irrelevant for excitation transfer at $\phi = 0$. At $\phi = 0$, one should thus consider the modulus of the subleading eigenvalue, $|\tilde{\mu}_{\rm PF}|$ [red dot in Fig. S2(e)], which is lower than the two local minima of $|\mu_{\rm PF}|$ when $\phi \neq 0$. We therefore conclude that the optimal phase for even $N$ is $\phi = 0$. EVEN N

## S3. DENSITY OF DARK STATES INCREASES WITH $\tau$

density

solve (10)

amplif. fator

more crossings = more solutions

We prove that the density of dark states—i.e., the number of dark states in a certain interval $[\phi, \phi + \Delta\phi]$ at fixed $\tau$—grows with $\tau$ in the region $\tau > \pi/2$. As discussed in the main text, outside of the degenerate case, dark states correspond to solutions of Eq. (10). Graphically, these solutions are the intersections of two functions, $y_1 = k\pi$ and $y_2 = 2\tau \sin[\pi(m-n)/N] \sin[\phi - \pi(m+n)/N]$. On the plane $(\phi, y)$, the function $y_1$ represents horizontal lines at integer multiples of $\pi$; at fixed $\tau, m, n, N$, the function $y_2$ is a sinusoidal function in $\phi$, with $\tau$ acting as an amplification factor, provided that $m, n$ satisfy $\sin[\pi(m-n)/N] \neq 0$. It is straightforward to see that the number of horizontal lines $y_1 = k\pi$ crossed by the function $y_2$ grows in any interval $[\phi, \phi + \Delta\phi]$ as $\tau$ increases, as was to be demonstrated. In other words, for larger values of $\tau$, a finer tuning on $\phi$ is needed to avoid dark states, solutions of $y_1 = y_2$. Searching for a robust transfer protocol, this further motivates our choice of restricting to $\tau < \tau^*$.

LARGER \tau requires a FINER tuning of \phi

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dark states?


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chirality